Flow Control Mechanism

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Introduction to Flow Control

Flow control is a critical mechanism in computer networks.

It dynamically regulates the pace at which data segments are transmitted from a sender to a receiver. This process is essential to prevent the receiver's buffer from overflowing and to ensure reliable data delivery. By constantly monitoring the receiver's capacity and feedback, flow control protocols adapt the transmission rate. Without effective flow control, an overly rapid data flow can lead to packet loss, retransmissions, and significant performance degradation, ultimately hindering overall network efficiency and stability.

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Why Do We Need Flow Control?

Varying Speeds

Different network devices have diverse processing speeds.

Buffer Capacity

Devices also have limited buffer capacities.

Prevent Data Loss

Flow control prevents data from being lost due to overload.

Avoid Retransmission

It reduces the need for costly data retransmissions.

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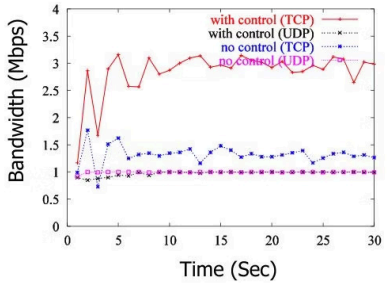
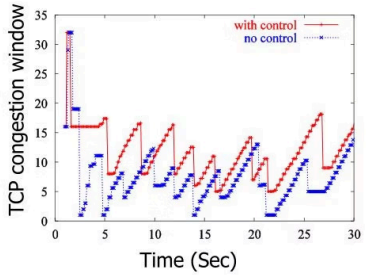
Flow Control





Flow control mechanism increases the bandwidth of adaptive flows by consuming bandwidth wasted by unresponsive flows

Types of Flow Control Mechanisms

Stop and Wait Protocol

Simple, but less efficient for high-speed networks.

Sliding Window Protocol

More efficient, allows multiple frames to be sent.

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Stop and Wait Protocol

The Stop and Wait Protocol is a fundamental data link layer flow control mechanism designed for reliable data transmission. At its core, the sender transmits a single data frame and then *stops* further transmission, *waiting* for an explicit acknowledgment (ACK) from the receiver. Only upon successful receipt of the ACK does the sender proceed to transmit the next data frame.

The step-by-step process is as follows:

1. The sender transmits a data frame and starts a timer.
2. The sender then pauses and waits for an acknowledgment (ACK) from the receiver.
3. If the receiver successfully receives the data frame, it sends an ACK back to the sender.
4. Upon receiving the ACK, the sender stops its timer and transmits the next data frame.
5. If the timer expires before an ACK is received (indicating a potential loss of the frame or ACK), the sender retransmits the original data frame and restarts the timer.

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Sliding Window Protocol

The Sliding Window protocol allows the sender to transmit multiple frames consecutively without waiting for an ACK after each one.

The receiver specifies a "window size," which dictates the maximum number of frames the sender can transmit before requiring an acknowledgment.

This significantly improves efficiency by keeping the communication channel busy.

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Flow Control vs. Error Control

Flow Control

Manages the rate of data transmission.

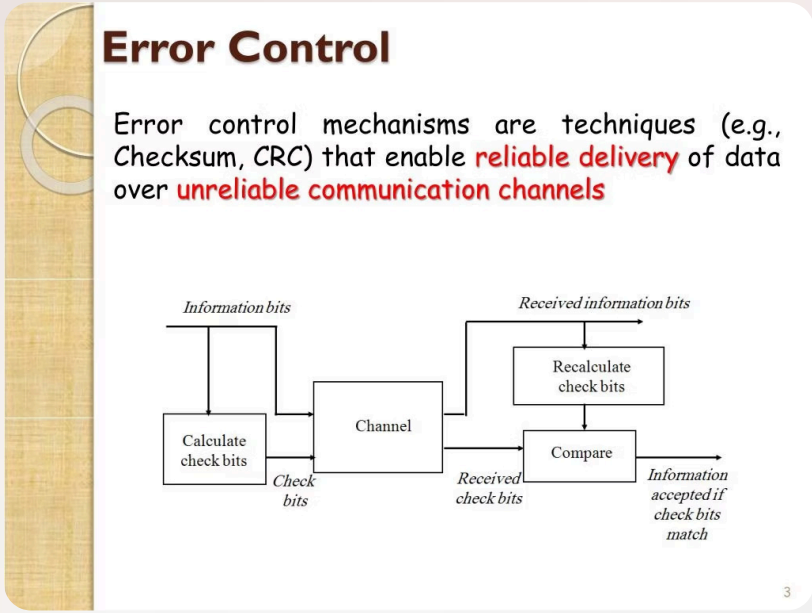
Prevents receiver overload.

Error Control

Ensures the accuracy and integrity of data delivery.

Detects and corrects errors during transmission.

Both mechanisms are crucial and work in tandem to achieve reliable data communication in computer networks.



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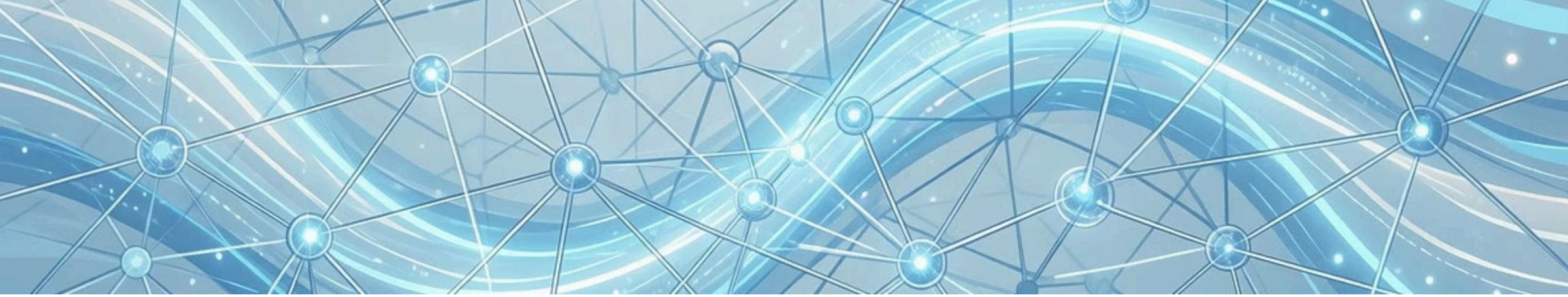

Conclusion

Essential Role

Flow control is vital for efficient and reliable data communication.

Performance Boost

Proper implementation enhances overall network performance.

Data Integrity

It actively prevents data loss and ensures smooth operation.

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References

- A. S. Tanenbaum and D. J. Wetherall, _Computer Networks_, 5th ed., Pearson Education, 2011.
- B. A. Forouzan, _Data Communications and Networking_, 5th ed., McGraw-Hill, 2013.

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